Review of Probability and Random Processes

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References

- Chapters 1 to 7 of the book: A. Papoulis, S. Unnikrishnan Pillai, “*Probability, Random Variables and Stochastic Processes*”, Fourth Edition, McGraw Hill, 2002.
- Chapters 1 to 5 of the book: H.Stark, J. Woods, “*Probability and Random Processes with Applications to Signal Processing*”, Third Edition, Pearson Education, 2002.
- Chapter 2 of the book: B. Hajek, ""*Random Processes for Engineers*“, Cambridge University Press, 2015. (available on the author's website: https://hajek.ece.illinois.edu/Papers/randomprocJuly14.pdf).

Probability Space

- Let $(\Omega, \mathcal{F}, P)$ be a probability space
- Ω: sample space
  - ❑ set of all possible outcomes of an experiment
- Each outcome (element of Ω) also called sample point
- $\mathcal{F}$: σ-field
  - ❑ collection of subsets of Ω
  - ❑ these subsets are called events
- P:
  - ❑ probability function
  - ❑ assigns a real number, $P(A)$, to each event $A \in \mathcal{F}$
- Axioms of probability:

I. $P(A) \geq 0$ for all $A \in \mathcal{F}$

II. $\mathrm{P}(\Omega) = 1$

III. If $A_1, A_2, A_3, \ldots \in \mathcal{F}$ are mutually exclusive, then:

- ❑ $P(\bigcup_{k=1}^{\infty} A_k) = \sum_{k=1}^{\infty} P(A_k)$

Continuity Property

- $P(.)$ is a continuous function
- Let $A_1, A_2, A_3, \ldots$ be events such that:

$$A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$$

- Also, let $A = \bigcup_{k=1}^{\infty} A_k$
- Then $P(A) = \lim_{n \to \infty} P(A_n)$
- Similarly, let $A_1, A_2, A_3, \ldots$ be events such that:

$$A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$$

- Also, let $A = \bigcap_{k=1}^{\infty} A_k$
- Then $P(A) = \lim_{n \to \infty} P(A_n)$

Example: Infinite Monkey Theorem

- **Theorem**: In an infinite sequence of symbols, each selected independently uniformly at random from some alphabet $\mathcal{A} = \{a_1, \dots, a_M\}$, any given sequence of symbols, say s, of finite length l appears with probability 1
  - *e.g.*, a monkey typing randomly on a keyboard for an infinite amount of time will type a given finite piece of text (*e.g.*, The Complete Works of Shakespeare) with probability 1

<u>Proof</u>:

- Divide the infinite sequence into blocks of l symbols each
- Let A be the event that the sequence s appears in at least one block; suffices to show that $P(A) = 1$
- Let A_n: the sequence s appears in at least one of the first n blocks
  - Note that $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$
- $P(A_n)$:
  - $1 - \left(1 - \left(\frac{1}{M}\right)^l\right)^n$
- A in terms of $A_1, A_2, A_3, \dots$:
  - $\bigcup_{k=1}^{\infty} A_k$
- $P(A) = \lim_{n \to \infty} P(A_n) = 1$ by continuity property

Conditional Probability

- $P(E \mid M) = \frac{P(E \cap M)}{P(M)}$ if $P(M) > 0$
- Suppose A_1, A_2, A_3 … form a partition of Ω and B is any event
- *Theorem of Total Probability*:
  - ❑ $P(B) = \sum_{i=1}^{\infty} P(B|A_i)P(A_i)$
- *Bayes' Rule*:
  - ❑ $\mathrm{P}(A_k|B) = \frac{P(B|A_k)P(A_k)}{\sum_{i=1}^{\infty} P(B|A_i)P(A_i)}$

Independence

- Two events A and B are independent if $P(A \cap B) = P(A)P(B)$
- Events $A_1, A_2, \dots, A_n$ are independent if for every $k \in \{2,3, \dots, n\}$ and indices $i_1, \dots, i_k$ such that $1 \leq i_1 < i_2 < \cdots < i_k \leq n$:
  - ❑ $P(A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}) = P(A_{i_1})P(A_{i_2}) \dots P(A_{i_k})$

Random Variables

- Let $(\Omega, \mathcal{F}, P)$ be a given probability space. A *random variable* X is a function from Ω to $\mathcal{R}$ that satisfies the following condition:
  - ❑ $\{\omega \in \Omega: X(\omega) \leq x\} \in \mathcal{F}$ for all $x \in \mathcal{R}$
- The function $F(x) = P(X \leq x),\ x \in \mathcal{R}$ called *Cumulative Distribution Function (CDF)* of X
- Properties of $F(x)$:

  1) $0 \leq F(x) \leq 1$ for all $x \in \mathcal{R}$

  2) If $x_1 < x_2$, then $F(x_1) \leq F(x_2)$

  3) $\lim_{x \to \infty} F(x) = 1$

  4) $\lim_{x \to -\infty} F(x) = 0$

Types of Random Variables

1) Discrete random variables
   - can be represented by *probability mass functions (PMFs)*
   - e.g., Bernoulli, Binomial, uniform, geometric, Poisson
2) Continuous random variables
   - X is a continuous random variable if its CDF, $F(x)$, is continuous and its derivative exists everywhere except at a countable set of points S
   - can be represented by *probability density functions (PDFs)*
   - PDF $f(x) \triangleq \begin{cases} \frac{dF(x)}{dx}, & x \notin S, \\ \text{any non−negative number}, & x \in S. \end{cases}$
   - e.g., uniform, Gaussian (normal), exponential, Rayleigh
3) Mixed random variables
   - can be represented by PDFs if impulse functions are allowed
   - e.g, $F(x) = \begin{cases} 1 - \frac{1}{2}e^{-x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$
   - PDF $f(x)$:
     - $\frac{1}{2}\delta(x) + \frac{1}{2}e^{-x}\,u(x)$
4) Other random variables
   - e.g., Cantor distribution

Conditional Probability Space

- Let Ω be a sample space and $\mathcal{F}$ be a σ-field of subsets of Ω
- Recall: probabilities $P(.)$ are any assignments of real numbers to the subsets in $\mathcal{F}$ that satisfy the axioms
- Given occurrence of event M with $\mathrm{P}(M) > 0$:

I. $P(A|M) \geq 0$ for all A

II. $P(\Omega|M) = 1$

III. If $A_1, A_2, A_3, \ldots$ are mutually exclusive, then:

- ❑ $P(\bigcup_{k=1}^{\infty} A_k|M) = \sum_{k=1}^{\infty} P(A_k|M)$

- Hence, $(\Omega, \mathcal{F}, P(.|M))$ *is a probability space*
  - ❑ called conditional probability space corresponding to event M

Conditional Probability Space (contd.)

- Results similar to those in original probability space $(\Omega, \mathcal{F}, P(.))$ hold in the conditional probability space $(\Omega, \mathcal{F}, P(.|M))$
- **E.g.**:
  - ❑ $\mathrm{P}(A \cup B|M) = P(A|M) + P(B|M) - P(A \cap B|M)$
  - ❑ If $A_1, A_2, A_3, \ldots$ are events s.t. $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ and $A = \bigcup_{k=1}^{\infty} A_k$, then $P(A|M) = \lim_{n\to\infty} P(A_n|M)$
- Often convenient to perform computations in conditional probability space

Conditional Distributions

- Let M be an event such that $P(M) > 0$
- Conditional CDF:
  - $F(x|M) = \frac{P(X \leq x, M)}{P(M)}$
- Conditional PDF:
  - $f(x|M) = \frac{dF(x|M)}{dx}$
- Conditional PMF:
  - $P(X = x_k|M) = \frac{P(X = x_k, M)}{P(M)}$
- If $A_1, A_2, A_3, \ldots$ form a partition of Ω, then:
  - $F(x) = \sum_{i=1}^{\infty} F(x|A_i)P(A_i)$
  - $f(x) = \sum_{i=1}^{\infty} f(x|A_i)P(A_i)$
- If X is a continuous random variable and A is an event, then:
  - $P(A|X = x) = \frac{f(x|A)P(A)}{f(x)}$
  - $P(A) = \int_{-\infty}^{\infty} P(A|X = x)f(x)dx$
  - $f(x|A) = \frac{P(A|X=x)f(x)}{\int_{-\infty}^{\infty} P(A|X=u)f(u)du}$

Example

- Coin with *unknown* probability of heads X
  - model as a *continuous* random variable with PDF $f(x)$
- A: the coin is tossed n times and a sequence s of heads and tails observed
  - e.g., $n = 5$; $s = HTHHT$
- Assume that in s, there are k H and $(n-k)$ T
- $f(x|A)$:
  - $\frac{x^k(1-x)^{n-k}f(x)}{\int_0^1 u^k(1-u)^{n-k}f(u)du}$
- **Note**: $P(A|X=x) = x^k(1-x)^{n-k}$, *not* $\binom{n}{k}x^k(1-x)^{n-k}$

Expectation and Variance

- X: discrete random variable with PMF:
  - $P(X = x_k) = p_k$ for $x_k \in S$
- $E(X) = \sum_{x_k \in S} p_k x_k$ if it exists
- X continuous with PDF $f_X(.)$
- $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$ if it exists
- Let $f_X(x) = \begin{cases} \frac{k}{x^2}, & x \geq 1, \\ 0, & \text{else.} \end{cases}$
  - where k is a normalizing constant
- $E(X)$:
  - ∞
- If $Y = -X$, then $E(Y)$:
  - $-\infty$
- Thus, $E(X)$ and $E(Y)$ do not exist
  - sometimes said to equal ∞ or $-\infty$ for convenience

Expectation and Variance (contd.)

- Let $f_X(x) = \frac{1}{\pi(1+x^2)}$
- $E(X)$:
  - ❑ of the form $\infty - \infty$
  - ❑ undefined
- In summary, expectation found using $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$ or $\sum_{x_k \in S} p_k x_k$ may be ∞ or $-\infty$ or undefined
- Necessary and sufficient conditions for above integral and summation to be well-defined:
  - ❑ $\int_{-\infty}^{\infty} |x| f_X(x) dx < \infty$ and $\sum_{x_k \in S} p_k |x_k| < \infty$, respectively
- Hence definitions on previous slide become:
  - ❑ $E(X) = \sum_{x_k \in S} p_k x_k$ if $\sum_{x_k \in S} p_k |x_k| < \infty$
  - ❑ $E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$ if $\int_{-\infty}^{\infty} |x| f_X(x) dx < \infty$
- If X discrete with PMF $P(X = x_k) = p_k$ for $x_k \in S$, then $E(g(X)) = \sum_{x_k \in S} p_k g(x_k)$
- If X continuous with PDF $f_X(.)$, then $E(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$
- $\mathrm{var}(X) = E\left(\left(X - E(X)\right)^2\right) = E(X^2) - E(X)^2$

Random Vectors

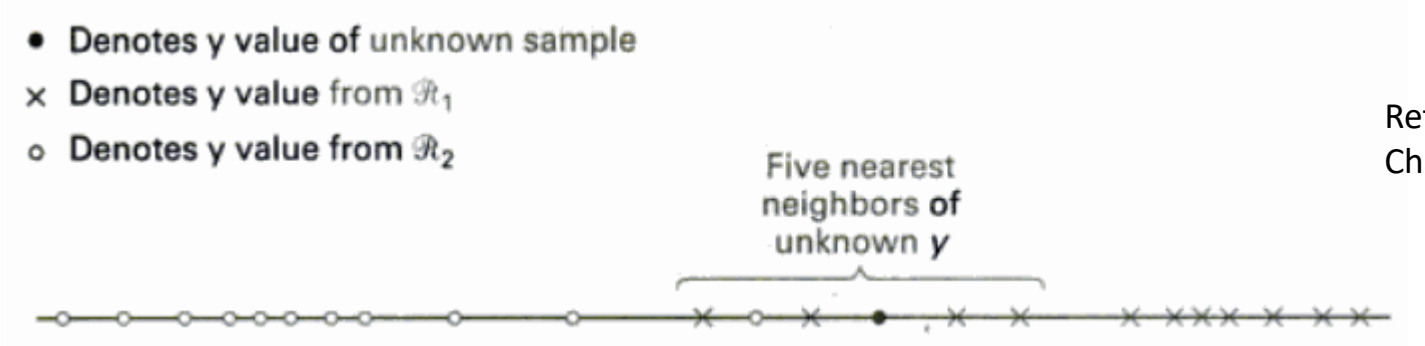
- Consider vector of n random variables:
  - $\boldsymbol{X} = (X_1, X_2, \dots, X_n)$
- Joint CDF:
  - $F_{X_1,\dots,X_n}(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$
- Joint PDF:
  - $f_{X_1,\dots,X_n}(x_1, \dots, x_n) = \frac{\partial^n F_{X_1,\dots,X_n}(x_1,\dots,x_n)}{\partial x_1 \dots \partial x_n}$
  - $P((X_1, X_2, \dots, X_n) \in D) = \int_D f_{X_1,\dots,X_n}(x_1, \dots, x_n) dx_1 \dots dx_n$
- Joint PMF:
  - $P(X_1 = x_1, \dots, X_n = x_n)$
- $X_1, \dots, X_n$ independent if the events $\{X_1 \in A_1\}, \dots, \{X_n \in A_n\}$ are independent for all $A_1 \subseteq \mathcal{R}$, ..., $A_n \subseteq \mathcal{R}$
- $X_1, \dots, X_n$ independent iff:
  1) $F_{X_1,\dots,X_n}(x_1, \dots, x_n) = F_{X_1}(x_1) \dots F_{X_n}(x_n)$
  2) $f_{X_1,\dots,X_n}(x_1, \dots, x_n) = f_{X_1}(x_1) \dots f_{X_n}(x_n)$ (continuous case)
  3) $P(X_1 = x_1, \dots, X_n = x_n) = P(X_1 = x_1) \dots P(X_n = x_n)$ (discrete case)
- If $g(X_1, \dots, X_n) = \sum_k a_k g_k(X_1, \dots, X_n)$, then:
  - $E\big(g(X_1, \dots, X_n)\big) = \sum_k a_k E\big(g_k(X_1, \dots, X_n)\big)$

Random Vectors (contd.)

- Mean Vector:
  - $\boldsymbol{\mu_X} = (\mu_1, \mu_2, \dots, \mu_n)$, where $\mu_j = E(X_j)$
- Covariance matrix:
  - $\mathrm{K}_{\boldsymbol{X}}$: $n \times n$ matrix with (i, j)'th element $C_{i,j} = \mathrm{Cov}(X_i, X_j)$
- Properties:
  - $\mathrm{K}_{\boldsymbol{X}}$ is a symmetric matrix
  - Diagonal elements of $\mathrm{K}_{\boldsymbol{X}}$ are variances of $X_1, X_2, \dots, X_n$
  - $\mathrm{K}_{\boldsymbol{X}} = E((\boldsymbol{X} - \boldsymbol{\mu_X})(\boldsymbol{X} - \boldsymbol{\mu_X})^T)$
  - For all i, j, ${C_{i,j}}^2 \leq C_{i,i} C_{j,j}$

Example: Automatic Recognition of Black-Lung Disease

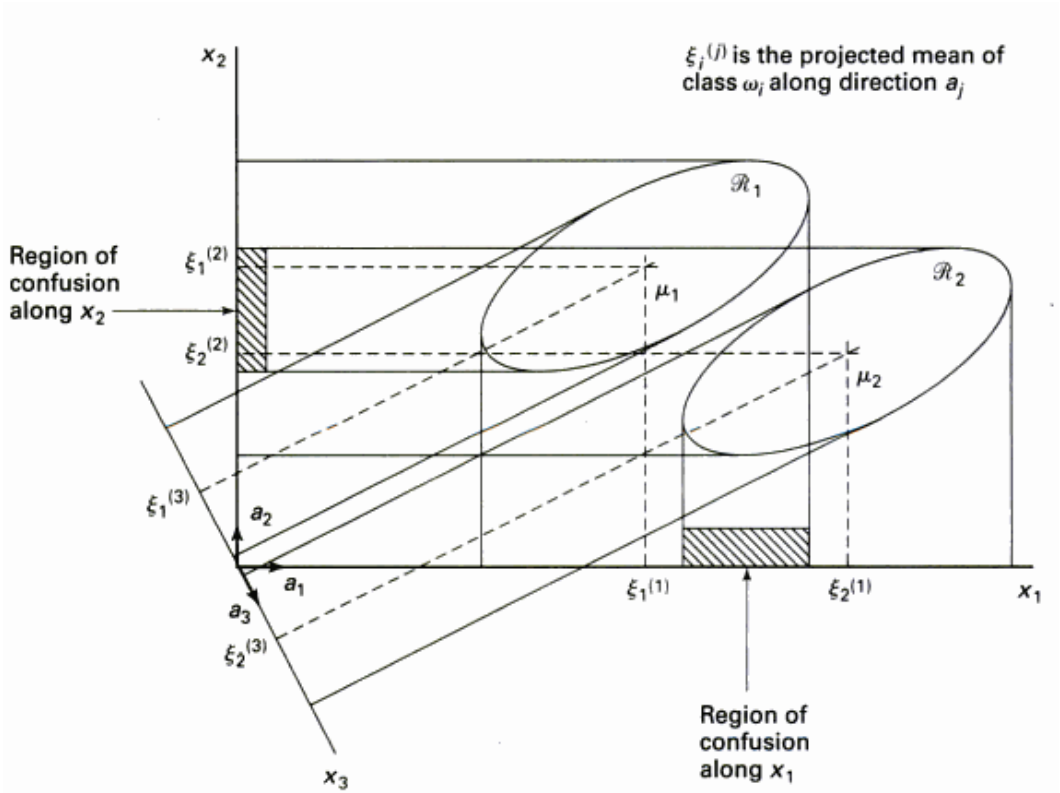
- n-component measurement vector $\boldsymbol{X}$ extracted from X-ray of patient
- Database of measurement vectors:
  - ❑ $\boldsymbol{X}'$ from people known to have disease, and
  - ❑ $\boldsymbol{X}''$ from healthy people available
- Want to find whether patient has disease
- Project $\boldsymbol{X}$ in direction of $\boldsymbol{a}$ to get a feature (scalar) $Y = \boldsymbol{a}^T\boldsymbol{X}$
  - ❑ also project vectors from database
- Use N-nearest-neighbor rule to make decision



Ref: Stark and Woods, Chapter 5

Example: Automatic Recognition of Black-Lung Disease

- For above rule to work well, *select direction of projection* $\boldsymbol{a}$ *so that r. vector projections of diseased and healthy people from database cluster in far-apart regions*
  - ❑ e.g., in figure, $\boldsymbol{a} = \boldsymbol{a_3}$ better than $\boldsymbol{a} = \boldsymbol{a_1}$ and $\boldsymbol{a} = \boldsymbol{a_2}$



Ref: Stark and Woods, Chapter 5

Example: Automatic Recognition of Black-Lung Disease

- Commonly used rule: select $\boldsymbol{a}$ that maximizes the function:
  - ❑ $J(\boldsymbol{a}) = \frac{(\eta_1 - \eta_2)^2}{\sigma_1^2 + \sigma_2^2}$, where
  - ❑ η_1, σ_1^2: mean and variance of $\boldsymbol{a}^T \boldsymbol{X}'$ (projection of $\boldsymbol{X}'$ on $\boldsymbol{a}$)
  - ❑ η_2, σ_2^2: mean and variance of $\boldsymbol{a}^T \boldsymbol{X}''$
  - ❑ can be found from *mean vectors* and *covariance matrices*
- Direction $\boldsymbol{a}$ that maximizes $J(\boldsymbol{a})$ can be found using matrix calculations
  - ❑ details omitted

Functions of Random Variables

- Let $Y = g(X)$
- Given $F_X(.)$, want $F_Y(.)$
- $F_Y(y) = P(g(X) \leq y)$
  - $= P\left(X \in B_y\right)$,
  - where $B_y = \{x: g(x) \leq y\}$
- Let X be continuous with PDF $f_X(.)$ and $Y = g(X)$
- Let $g(.)$ be such that Y is continuous
- Suppose $y_1 = g(x)$ has finite or countably infinite number of roots
- Then:
  - $f_Y(y_1) = \sum_{x \in S} \frac{f_X(x)}{|g'(x)|}$,
  - where S is the set of roots

Functions of Random Variables (contd.)

- Let $Z = g(X, Y)$ and $W = h(X, Y)$
- Then $F_{Z,W}(z, w) = P(g(X, Y) \leq z, h(X, Y) \leq w)$
- So, $F_{Z,W}(z, w) = P((X, Y) \in D_{z,w})$,
  - where $D_{z,w} = \{(x, y): g(x, y) \leq z, h(x, y) \leq w\}$
- $P((X, Y) \in D_{z,w})$ can be found from joint PDF or joint PMF of X and Y
- When the system $z_1 = g(x, y)$, $w_1 = h(x, y)$ has multiple roots:
  - $f_{Z,W}(z_1, w_1) = \sum_{(x,y)\in S} \frac{f_{X,Y}(x,y)}{|\mathrm{J}(x,y)|}$
  - where S is the set of roots and $\mathrm{J}(x, y) = \begin{vmatrix} \frac{\partial g(x,y)}{\partial x} & \frac{\partial g(x,y)}{\partial y} \\ \frac{\partial h(x,y)}{\partial x} & \frac{\partial h(x,y)}{\partial y} \end{vmatrix}$

Functions of Random Variables (contd.)

- Let $Y = g(X_1, \dots, X_n)$
- $F_Y(y)$:
  - $P(g(X_1, \dots, X_n) \leq y)$
  - $P((X_1, \dots, X_n) \in D)$, where $D = \{(x_1, \dots, x_n) \in \mathcal{R}^n : g(x_1, \dots, x_n) \leq y\}$
- Consider the system $Y_i = g_i(X_1, \dots, X_n), i = 1, \dots, n$
- Has a single root at $(y_1, \dots, y_n)$
- $f_{Y_1,\dots,Y_n}(y_1, \dots, y_n)$:
  - $\frac{f_{X_1,\dots,X_n}(x_1,\dots,x_n)}{|J(x_1,\dots,x_n)|}$
  - where $J(x_1, \dots, x_n) = \begin{vmatrix} \frac{\partial g_1}{\partial x_1} & \dots & \frac{\partial g_1}{\partial x_n} \\ \dots & \dots & \dots \\ \frac{\partial g_n}{\partial x_1} & \dots & \frac{\partial g_n}{\partial x_n} \end{vmatrix}$
- Multiple roots case: summation over roots

Example

- X, Y independent, each $N(0,1)$
- Want $P((X,Y) \in D)$ where D is unit disc
- Change to polar coordinates:
  - $R = \sqrt{X^2 + Y^2}$ and $\Theta = \tan^{-1}\left(\frac{Y}{X}\right)$
- $f_{R,\Theta}(r,\theta)$:
  - $\frac{r}{2\pi} e^{\frac{-r^2}{2}}$ for $r \geq 0, \theta \in [0,2\pi]$
- Integrate to get $P\big((X,Y) \in D\big) = 1 - e^{-1/2}$
- $f_R(r)$:
  - $re^{\frac{-r^2}{2}}$ for $r \geq 0$
- $f_\Theta(\theta)$:
  - $\frac{1}{2\pi}$ for $\theta \in [0,2\pi]$
- Thus, R is Rayleigh distributed, Θ is uniformly distributed and they are independent

Inequalities

- Markov's inequality:
  - ❑ Let $X \geq 0$ and $a > 0$. Then:

    1) $P(X \geq a) \leq \frac{E(X)}{a}$
- Chebyshev's inequality:
  - ❑ Let X be arbitrary and $\epsilon > 0$. Then:

    2) $P(|X - E(X)| \geq \epsilon) \leq \frac{\mathrm{var}(X)}{\epsilon^2}$
- For non-negative r.v. X and $a > 0$, want upper bounds on $P(X \geq a)$ that are tighter than that in 1)
- For $x \geq 0$, let $f(x)$ be any non-negative valued, strictly increasing function
- $P(X \geq a) = P(f(X) \geq f(a))$

3) So, $P(X \geq a) \leq \frac{E(f(X))}{f(a)}$

- 3) provides a family of inequalities
  - ❑ e.g., $P(X \geq a) \leq \frac{E(\log(X+1))}{\log(a+1)}$ or $\frac{E(X^n)}{a^n}$, where $n > 0$
- A useful choice: $f(x) = e^{tx}$, where $t > 0$

4) $P(X \geq a) \leq \frac{E(e^{tX})}{e^{ta}}$

  - ❑ Chernoff bound

Characteristic Functions

- Characteristic function of X:
  - $\phi_X(\omega) = E(e^{j\omega X})$
- If X has PDF $f_X(x)$, then:
  - $\phi_X(\omega) = \int_{-\infty}^{\infty} e^{j\omega x} f_X(x) dx$
- Relation to Fourier transform of $f_X(x)$:
  - $\phi_X(\omega) = F(-\omega)$
- Inversion formula:
  - $f_X(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \phi_X(\omega) e^{-j\omega x} d\omega$
- If X has PMF $P(X = x_i) = p_i$, then:
  - $\phi_X(\omega) = \sum_i e^{j\omega x_i} p_i$

Existence and Uniqueness

- The characteristic function of a random variable X *always exists*
- X and Y have the same characteristic function iff they have the same distribution:
  - $\phi_X(\omega) = \phi_Y(\omega)\ \forall\omega$ iff $F_X(x) = F_Y(x)\ \forall x$
- Thus, if characteristic function of X known, then its distribution can be found via tables

Sum of Independent Random Variables

- X, Y independent and $Z = X + Y$
- $\phi_Z(\omega)$:
  - $\phi_X(\omega)\phi_Y(\omega)$
- Can be shown that $f_Z(z) = f_X(z) * f_Y(z)$
- Above result similar to $y(t) = x(t) * h(t) \Leftrightarrow Y(\omega) = X(\omega)H(\omega)$

Moments

- $E(X^n)$: n'th moment of X
  - $n \geq 1$, integer
- If X has PDF $f_X(x)$, then $E(X^n)$:
  - $\int_{-\infty}^{\infty} x^n f_X(x) dx$
- If X has PMF $P(X = x_i) = p_i$, then $E(X^n)$:
  - $\sum_i {x_i}^n p_i$

Moment Generating Functions

- Moment Generating Function of X:
  - ❑ $\Lambda_X(s) = E(e^{sX})$
- Relation to Laplace transform of $f_X(x)$:
  - ❑ $\Lambda_X(s) = L(-s)$
- MGFs and CFs have similar properties
- E.g., if X, Y independent and $Z = X + Y$, then:
  - ❑ $\phi_Z(\omega) = \phi_X(\omega)\phi_Y(\omega)$ and $\Lambda_Z(s) = \Lambda_X(s)\Lambda_Y(s)$

Finding Moments

- Recall $E(X^n)$: n'th moment of X
- $\Lambda_X(s) = \int_{-\infty}^{\infty} e^{sx} f_X(x)dx$
- Differentiate w.r.t s:
  - ❑ $\Lambda_X{}'(s) = \int_{-\infty}^{\infty} e^{sx} x f_X(x)dx$
  - ❑ So, $\Lambda_X{}'(0) = E(X)$
- $\Lambda_X{}''(s)$:
  - ❑ $\int_{-\infty}^{\infty} e^{sx} x^2 f_X(x)dx$
  - ❑ So, $\Lambda_X{}''(0) = E(X^2)$
- Similarly, $\Lambda_X{}^{(n)}(s) = \int_{-\infty}^{\infty} e^{sx} x^n f_X(x)dx$
  - ❑ So, $\Lambda_X{}^{(n)}(0) = E(X^n)$

Joint Characteristic Functions

- Joint Characteristic Function of X and Y:
  - $\Phi_{X,Y}(\omega_1, \omega_2) = E\left(e^{j(\omega_1 X + \omega_2 Y)}\right)$
- If joint PDF is $f_{X,Y}(x, y)$, then:
  - $\Phi_{X,Y}(\omega_1, \omega_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{j(\omega_1 x + \omega_2 y)} f_{X,Y}(x, y) dx dy$
  - 2D Fourier transform with a sign change
- Inversion formula:
  - $f_{X,Y}(x, y) = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \Phi_{X,Y}(\omega_1, \omega_2) e^{-j(\omega_1 x + \omega_2 y)} d\omega_1 d\omega_2$
- $\Phi_{X,Y}(\omega_1, \omega_2)$ always exists; (X, Y) and (Z, W) have same joint characteristic function iff they have same joint CDF
  - can use tables to obtain joint distribution corresponding to given joint characteristic function

Properties

1) $\Phi_X(\omega) = \Phi_{X,Y}(\omega, 0)$ and $\Phi_Y(\omega) = \Phi_{X,Y}(0, \omega)$

2) If X, Y independent, then $\Phi_{X,Y}(\omega_1, \omega_2)$:
   - $\Phi_X(\omega_1)\Phi_Y(\omega_2)$

3) Is the converse of property (2) true?
   - Yes, by inversion formula

Joint Moments

- $E(X^n Y^m), n \geq 1, m \geq 1$
  - $(n'm)$'th joint moment
- Joint moment generating function:
  - $\Lambda_{X,Y}(s_1, s_2) = E(e^{s_1 X + s_2 Y})$
- Finding $E(X^n Y^m)$ from $\Lambda_{X,Y}(s_1, s_2)$:
  - $E(X^n Y^m) = \frac{\partial^{n+m} \Lambda_{X,Y}(0,0)}{\partial {s_1}^n \partial {s_2}^m}$
- Similarly, $E(X^n Y^m) = \frac{1}{j^{n+m}} \frac{\partial^{n+m} \Phi_{X,Y}(0,0)}{\partial {\omega_1}^n \partial {\omega_2}^m}$

Characteristic Functions of Random Vectors

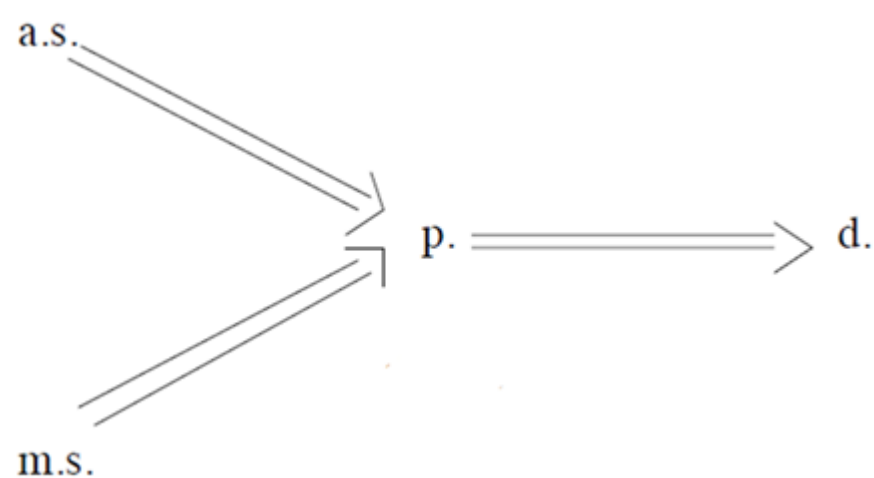
- Characteristic function of $\boldsymbol{X} = (X_1, X_2, \dots, X_n)$:
  - $\Phi_{\boldsymbol{X}}(\boldsymbol{\omega}) = E\left(e^{j(\boldsymbol{\omega}^T \boldsymbol{X})}\right)$
  - where $\boldsymbol{\omega} = (\omega_1, \dots, \omega_n)$
  - n-dimensional Fourier transform with a sign change
- $\Phi_{\boldsymbol{X}}(\boldsymbol{\omega})$ always exists; $\boldsymbol{X}$ and $\boldsymbol{Y}$ have same characteristic function iff they have same joint distribution
- If $X_1, X_2, \dots, X_n$ independent, then:
  - $\Phi_{\boldsymbol{X}}(\boldsymbol{\omega}) = \prod_{i=1}^{n} \Phi_{X_i}(\omega_i)$
- Converse also true
- $E\left({X_1}^{k_1} \dots {X_n}^{k_n}\right)$:
  - $\frac{1}{j^{k_1+\cdots+k_n}} \frac{\partial^{k_1+\cdots+k_n} \Phi_{\boldsymbol{X}}(0,\dots,0)}{\partial {\omega_1}^{k_1} \dots \partial {\omega_n}^{k_n}}$
- Moment generating function of $\boldsymbol{X} = (X_1, X_2, \dots, X_n)$:
  - $\Lambda_{\boldsymbol{X}}(\boldsymbol{s}) = E\left(e^{\boldsymbol{s}^T \boldsymbol{X}}\right)$
  - where $\boldsymbol{s} = (s_1, \dots, s_n)$

Example

- Suppose $X_1, \dots, X_n$ independent and X_i is Poisson(λ_i)
- Want the distribution of $X = X_1 + \cdots + X_n$
- $\Phi_X(\omega)$:
  - $\prod_{i=1}^{n} \Phi_{X_i}(\omega)$
  - $\exp\left((\lambda_1 + \dots + \lambda_n)(e^{j\omega} - 1)\right)$
- Hence, $X = X_1 + \cdots + X_n$ is Poisson$(\lambda_1 + \dots + \lambda_n)$

Convergence of Random Sequences

- $X_1, X_2, X_3, \ldots$ converges *almost surely* to X if $\lim_{n\to\infty} X_n(\omega) = X(\omega)$ for every $\omega \in A$, where $A \subseteq \Omega$ and $P(A) = 1$
- $X_1, X_2, X_3, \ldots$ converges to X *in probability* if for every $\epsilon > 0$, $\lim_{n\to\infty} P(\{\omega: |X_n(\omega) - X(\omega)| \geq \epsilon\}) = 0$
- Let $E\left(X_n{}^2\right) < \infty$ for all $n = 1,2,3, \ldots$ and $E(X^2) < \infty$. $X_1, X_2, X_3, \ldots$ converges to X *in mean square sense* if $\lim_{n\to\infty} E(|X_n - X|^2) = 0$
- $X_1, X_2, X_3, \ldots$ converges *in distribution* to X if $\lim_{n\to\infty} F_n(x) = F(x)$ for all points $x \in \mathcal{R}$ such that $F(x)$ is continuous at x



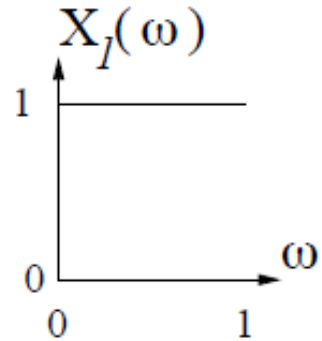
Ref: Chapter 2, Hajek

Example

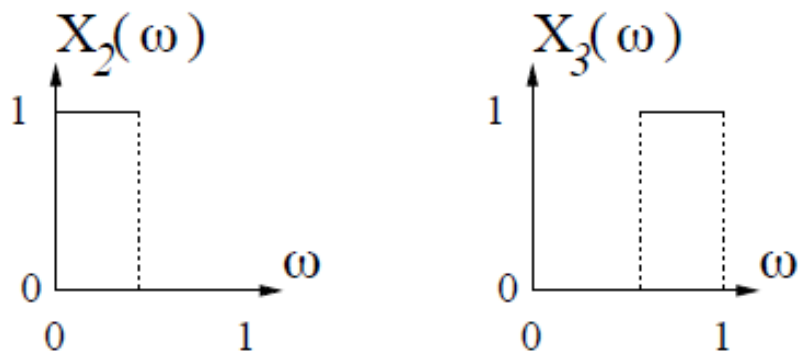
- $\Omega = [0,1], P([a,b]) = b - a,\ 0 \le a \le b \le 1$
- $X_n(\omega) = \begin{cases} \sqrt{n}, & 0 \le \omega < \frac{1}{n}, \\ 0, & \text{else.} \end{cases}$
- Convergence in probability?
  - ❑ yes, to 0
- Almost sure convergence?
  - ❑ yes, to 0
- Convergence in mean square sense?
  - ❑no

Example

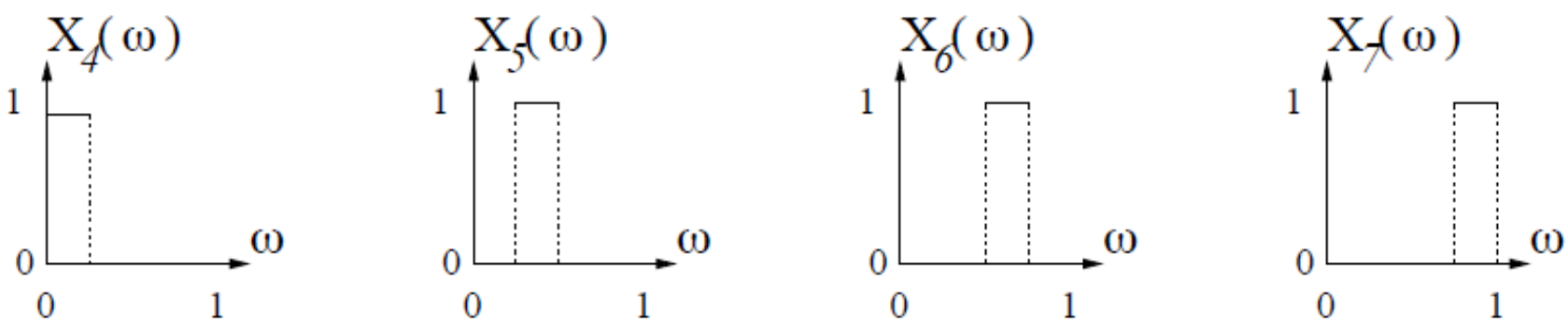
- $\Omega = [0,1], P([a,b]) = b - a,\ 0 \le a \le b \le 1$



- $X_1, X_2, X_3, \ldots$ converges in m.s. sense to 0, but does not converge a.s.



Ref: Hajek, Chapter 2



Laws of Large Numbers (LLNs)

1) *Weak LLN*: Let $X_1, X_2, X_3, \ldots$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Then $\frac{X_1+\cdots+X_n}{n}$ converges to μ in probability.
2) *Mean Square Version of LLN*: Let $X_1, X_2, X_3, \ldots$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Then $\frac{X_1+\cdots+X_n}{n}$ converges to μ in m.s.

- In fact, the convergence in 1) and 2) holds even if $X_1, X_2, X_3, \ldots$ are pairwise uncorrelated with the same mean μ and $\mathrm{var}(X_i) \leq \sigma^2 < \infty$ for all i

3) *Strong LLN*: Let $X_1, X_2, X_3, \ldots$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Then $\frac{X_1+\cdots+X_n}{n}$ converges to μ almost surely.

Example

- Casino games always designed so that casino makes profit in long run although it may make losses in short-term
- $p = P(\text{casino wins a game}) > \frac{1}{2}$
- Sequence of games played at a casino (possibly by different gamblers)
- X_i: casino's gain in game i
  - $\begin{cases} 1, & \text{if casino wins,} \\ -1, & \text{else.} \end{cases}$
- $X_1, X_2, X_3, \ldots$ i.i.d. with mean:
  - $\mu = 2p - 1$
- For large n, $X_1 + \cdots + X_n$:
  - $\approx \mu n = (2p - 1)n$
  - a large positive number

Central Limit Theorem (CLT)

- Let $S_n = X_1 + \cdots + X_n$
- *CLT*: If $X_1, X_2, X_3, \ldots$ i.i.d. with mean μ and variance $\sigma^2 < \infty$, then $\frac{S_n - n\mu}{\sqrt{n}\sigma}$ converges in distribution to a standard normal random variable

Example

- Astronomer wants to measure distance, d, in light-years from their observatory to a distant star
- Takes n measurements $X_1, \ldots, X_n$ and finds average
  - i.i.d. random variables with mean d and variance 4
- Want n such that estimate accurate to within 0.5 light-years with probability at least 0.95
- By CLT, select smallest n for which $P\left(\frac{-\sqrt{n}}{4} \leq Z \leq \frac{\sqrt{n}}{4}\right) \geq 0.95$, where Z is a standard normal random variable
- From table of normal distribution, $n = 62$